



Release Report

Frontend · UI · Backend v3.1

From mockup to production: bento dashboard,
real Sepolia transactions, team email invitations,
QSAFETY scoring and quantum-safe operating model.

Issued: May 10, 2026

Project: github.com/Quantum-Qustody/appv3

Production: <https://www.quantumqustody.com>

01 · EXECUTIVE SUMMARY

From mockup to production

This release takes Quantum Qustody from a static design exploration to a working production application: a real institutional sandbox at <https://www.quantumqustody.com> with a Supabase backend, Vercel frontend, real Ethereum Sepolia transactions via MetaMask, team email invitations, and a redesigned bento-style dashboard. Every page is responsive, every action writes to the database, and every transaction emits real-time updates.

Stack	Frontend	Backend	Chain
	React 18 · Vite 6	Supabase · PostgreSQL	Ethereum Sepolia
Hosting	Vercel (auto-deploy main)	Supabase Cloud	Public RPC
Auth	Supabase Auth (email)	RLS off (sandbox)	MetaMask EIP-1193
Domain	quantumqustody.com	jelyszovakrmwnjplphz.supabase.co	sepolia.etherscan.io

02 · BY THE NUMBERS

Scope of this release

Metric	Description
1,599	lines of React in App.jsx
6	Supabase migrations applied
1	Supabase Edge Function deployed
11	in-app pages (sidebar)
8	blockchain networks seeded
3	subscription plans (Starter / Pro / Enterprise)
4	Sepolia faucets linked from the UI
20	Vercel review items addressed

03 · FRONTEND & UI

Landing page

- Removed underscore _ after QUANTUM_QUSTODY in nav and footer
- Removed APPROACH / DIFFERENCE / SCENARIOS / ENGAGEMENT internal anchor links — single HOME nav
- Removed MVP ALPHA SANDBOX hero pill and SANDBOX badges everywhere
- Footer now has three official social icons: **X** · **Instagram** · **LinkedIn**, all linking to *@quantumqustody*
- QQ brand mark added to nav (top-left), footer, auth screen, and setup wizard
- Hero scaled responsively: text-4xl sm:text-5xl md:text-7xl
- Difference table wrapped in horizontal scroll for narrow screens

In-app shell

- Brand mark + transparent QQ logo in the top header (drop-shadow purple glow)
- **Quantum Safety Score atom** — animated SVG ring, live computed from PQC posture, control model, threshold settings, active team, connected wallets, and persisted settings (0–100, four tiers)
- Mobile drawer sidebar with hamburger toggle, audit log accessible via right-edge button
- Slide-in / slide-out animations (CSS keyframes added to index.css)
- Dark and Light themes via CSS variables, persisted in user_settings
- Audit log right-rail with real-time INSERT subscription on audit_logs

Sidebar (final order)

- DASHBOARD — bento layout, hero cards, stat row, recent activity
- DIGITAL ASSETS — was Asset Boundary; live MetaMask import + manual
- IMPORT BANK **SOON** — open-banking pages, locked
- GOVERNED MOVEMENT — Send / Swap / Bridge on Sepolia
- OVERVIEW — 8-card live counts (txs, team, assets, wallets, banks...)
- TEAM — email invitations, role cards, threshold settings
- EVIDENCE VIEWER — TX history, accounting CSV/JSON export
- HOW IT WORKS — 10-step journey + Quantum Safety module
- SUPPORT — ticket form persisting to support_tickets
- BILLING — plans, payment methods, invoices
- SETTINGS — language + theme + context + control + evidence

04 · DASHBOARD (BENTO)

Hero, actions, stats, activity

Replaced scenario-heavy hub with a real institutional dashboard. Bento grid: TOTAL_VALUE hero (2/3) + QSAFETY (1/3), then three big SEND / SWAP / BRIDGE action cards that route to Governed Movement, then two setup cards (IMPORT_CRYPTO + IMPORT_BANK with COMING_SOON sticker), a 5-up live stat row, and a final row with Recent Activity + Quick Links.

Card	Source	Real-time
TOTAL_VALUE	assets + banks (USD sum)	yes — recomputed on data change
QSAFETY atom	computed from org/threshold/team/wallets	yes — live
SEND / SWAP / BRIDGE	static actions	click → Governed Movement
IMPORT_CRYPTO	Wallet flow	click → MetaMask connect
IMPORT_BANK	Locked (Coming Soon sticker)	—
Stat row	live counts (banks, wallets, team, threshold, trust)	yes
Recent Activity	audit_logs · last 8	yes — Supabase realtime
Quick Links	team / evidence / how-it-works / settings	—

05 · BLOCKCHAIN INTEGRATION

Real Sepolia transactions

Governed Movement now executes on Ethereum Sepolia testnet through MetaMask. Connect button uses EIP-1193, auto-switches networks via `wallet_switchEthereumChain`, reads live balances from a public RPC, and signs transactions locally. Quantum Qustody **never receives a private key** — the signer remains with the user's wallet.

Library

ethers v6 (BrowserProvider, JsonRpcProvider, Contract)

Helpers (src/sepolia.js)

- **useWallet()** hook — connect, disconnect, ensureSepolia, refreshBalance, sendEth, wrapEth, unwrapWeth
- **readBalance(addr)** — public RPC balance read
- **readWethBalance(addr)** — Sepolia WETH9 balanceOf
- **FAUCETS** — Google Cloud / Alchemy / QuickNode / Infura quick links

Action mapping

Action	On-chain?	Implementation
Connect Wallet	Real	<code>window.ethereum.request({method:'eth_requestAccounts'})</code>
Read Balance	Real	Public Sepolia RPC
SEND ETH	Real	<code>signer.sendTransaction({to, value: parseEther(...)})</code>
SWAP (ETH→WETH)	Real	WETH9 deposit() at <code>0xFF9976782d46CC05630D1f6eBAb18b2324d6B14</code>
BRIDGE	Demo	Loopback transfer until cross-chain SDK integrated

Each confirmed transaction persists to `movement_requests` with a `tx_hash` in `step_data`, appears in the Evidence Viewer's TRANSACTIONS tab, and exports to CSV/JSON. Etherscan links are clickable on every row.

06 · TEAM INVITATIONS

Email-based onboarding

Team members are now invited by email through Supabase Auth's `admin.inviteUserByEmail`. The invitee receives a magic-link, sets a password, and is automatically attached to the inviter's organisation as an active participants row with the role chosen at invite time.

End-to-end flow

Step	Where	What
1	Frontend	Admin opens Team → INVITE_BY_EMAIL → fills email, role, function
2	Edge Function	team-invite inserts pending invite + calls inviteUserByEmail
3	Supabase Auth	Sends an invitation email containing the redirect URL <code>?invite=TOKEN</code>
4	Invitee	Clicks link, sets password, lands on app at <code>https://www.quantumqustody.com/?invite=TOKEN</code>
5	Trigger	<code>on_auth_user_created_invite</code> fires <code>accept_team_invitation</code>
6	Fallback	If trigger missed (existing user), client RPC retries on sign-in
7	Realtime	Admin's Team page renders the new ACTIVE MEMBER without refresh

Resilience

- Trigger functions wrapped in EXCEPTION blocks — `auth.users` INSERT never rolls back
- `invitation_link_failures` table captures any caught error with reason for debugging
- `institution_fn` made nullable so an invitation without a function doesn't break enum cast
- Empty-string `institution_fn` is normalised to null before insert
- Existing-user invites fall back to `signInWithOtp` with same metadata
- URL fallback re-invokes RPC if the database trigger missed the link (covers existing users)
- Real-time subscription on participants and `team_invitations` tables — admin sees acceptance live

07 · BACKEND & SCHEMA

Supabase migrations applied

Migration	Purpose
002_review_changes.sql	New tables: banks · chains · wallets · support_tickets · plans · org_subscriptions · invoices · p
003_team_invitations.sql	team_invitations table with token + expiry. accept_team_invitation RPC. on_auth_user_create
004_no_mock_seed.sql	Wipes mock participants and assets seeded by scenario-engine. Adds BEFORE INSERT trig
005_invitation_resilience.sql	Wraps trigger functions in EXCEPTION handlers. invitation_link_failures debug table. Defensi
006_invitation_function_nullable.sql	alter table participants alter column institution_fn drop not null. Hardened RPC handles empty

Edge function — team-invite

Path: supabase/functions/team-invite/index.ts

- **action: send** — insert pending invite, call inviteUserByEmail, fall back to signInWithOtp for existing users
- **action: resend** — re-trigger Supabase invitation email for same token
- **action: revoke** — flip status to revoked
- **action: list** — return pending/accepted/revoked rows for the org
- Reads caller JWT to record invited_by
- CORS open; service-role key used internally; SITE_URL env var drives redirect

Realtime channels

- **audit-{session.id}** — INSERT on audit_logs
- **progress-{session.id}** — * on scenario_progress
- **participants-{org.id}** — * on participants
- **invitations-{org.id}** — * on team_invitations

08 · QUANTUM SAFETY MODULE

Why we are quantum safe

A dedicated section in How It Works explains the operating model in four moving parts. The button on the dashboard's QSAFETY card scrolls directly to it.

A · EOA Delegation

- We never receive or store the private key.
- Institution's signer (HSM, MPC, hardware) remains the root of trust.
- Only an EOA address is delegated under our policy.

B · ZK Selective Verification

- Every governance proof is a zero-knowledge attestation.
- Threshold met, policy applied, control passed — verified mathematically.
- Auditors verify without seeing balances, addresses, or policy text.

C · PQC Key Regeneration

- Delegated EOA's signing context rotates under ML-DSA-65 / ML-KEM-768.
- The rotation is itself ZK-proven for external verifiability.
- Old keys retire, new keys come online, policy chain never breaks.

D · Delegation Loop

- Authorise → Execute → Attest → Re-key, continuous.
- Every cycle re-evaluates policy, threshold, and trust posture.
- No static delegation — the delegation itself is governed.

QSAFETY Score formula

Computed live in the browser from real state: PQC posture (+30), control model (threshold +20 / committee +25 / single +8), threshold ≥ 2 (+10), active members (up to +15), connected wallets (up to +10), persisted settings (+5), eval objective (+5), completed scenarios (+5). Capped at 100. Tier labels:

STRONG ≥ 80 , GUARDED ≥ 60 , MODERATE ≥ 40 , AT-RISK < 40 .

09 · MOCK DATA REMOVAL

Real-data only

Every page now reads from Supabase exclusively. The hardcoded participants (Alexandra Chen, Marcus Webb, Diana Frost, Raj Patel, Sarah Liu) and asset rows (BTC Institutional Vault, ETH Treasury Reserve, USDC Liquidity Pool, SOL Operations Fund) — previously seeded by the scenario-engine edge function on each create_session — are gone.

How

- Migration 004 deletes existing rows by exact name match
- Two BEFORE INSERT triggers on participants and assets discard any future seed inserts
- Setup wizard role list replaced with abstract role descriptions (no fake names)
- Governed Movement actor labels read from real participants table (Requester / Approver) instead of hardcoded function names
- Team form's institution_fn defaults to empty (— Select —) instead of pre-filled mock value

Mobile responsiveness

Breakpoint	Behaviour
< 640 px (sm-)	Hero shrinks to text-4xl. QSAFETY label hides; ring-only atom. Hamburger drawer for sidebar.
640–767 (sm)	Stat row collapses to 2 cols. Forms in single column.
768–1023 (md)	Sidebar visible inline. Audit log still drawer.
≥ 1024 (lg)	Full layout: sidebar + content + audit log persistent rails.

11 · SECURITY & HYGIENE

What we never store, what we always store

Asset	Stored?
User private keys	Never — stays in MetaMask / hardware wallet
Wallet seed phrases	Never — not asked, not transmitted
Account passwords	Never plaintext — handled by Supabase Auth (bcrypt)
Bank account numbers	Last 4 only when import flow ships; rest tokenised by open-banking provider
Card numbers	Last 4 + brand only
Transaction hashes	Stored — public on-chain anyway
Wallet addresses	Stored — public
Email addresses	Stored — required for invites
Audit logs	Stored — institutional evidence

12 · DEPLOYMENT

How releases ship

GitHub main branch is the production source of truth. Vercel auto-builds on every push (~60 s).

Vercel

- Repo: github.com/Quantum-Qustody/appv3
- Framework auto-detect: Vite
- Build: `npm run build → dist/`
- Required env vars: `VITE_SUPABASE_URL`, `VITE_SUPABASE_ANON_KEY`
- Domain: www.quantumqustody.com

Supabase

- Project: `jelyszovakrmwnjplphz`
- Migrations live in `supabase-migrations/`
- Edge Functions: `team-invite` (deployed, `--no-verify-jwt`)
- Edge Function secret: `SITE_URL = https://www.quantumqustody.com`
- Auth → URL Configuration: `site_url` + redirect URLs whitelist

Companion documents (root of repo)

- **DEPLOY.md** — full deploy + clean-users + Sepolia test guide
- **INVITATIONS.md** — `team-invite` setup, SMTP, troubleshooting
- **README.md** — project root
- **supabase-migrations/00*.sql** — versioned schema migrations

13 · VERIFICATION CHECKLIST

Smoke test the live build

#	Action	Expected
1	Open https://www.quantumqustody.com in incognito	Clean nav with QQ logo, only HOME + ACCESS SANDBOX
2	Click X / Instagram / LinkedIn icon	Opens @quantumqustody on each platform
3	ACCESS SANDBOX → sign up new email	Sandbox setup wizard appears
4	Walk wizard → LAUNCH_SANDBOX	Lands on bento Dashboard
5	QSAFETY atom in header	Live ring + score, animated
6	WHY QUANTUM SAFE? button	How It Works → scrolls to module
7	DIGITAL ASSETS → IMPORT_CRYPTO	MetaMask prompt, Sepolia balance reads
8	Use a faucet, return, SEND 0.0001 to self	MetaMask signs, tx confirms on Sepolia
9	EVIDENCE VIEWER → Transactions	Row appears with status COMPLETED + Etherscan link
10	EXPORT_CSV	File downloads
11	TEAM → INVITE_BY_EMAIL	Invitation email sent, pending card visible
12	Invitee opens link, signs in	Auto-linked, admin sees ACTIVE MEMBER live
13	Mobile (Cmd+Shift+M, iPhone SE)	Hamburger sidebar, hero shrinks, atoms readable
14	SETTINGS → Theme: Light	UI flips, persisted in user_settings
15	SUPPORT → submit ticket	Row in support_tickets table

14 · WHAT'S NEXT

Roadmap candidates

- **WalletConnect (Reown AppKit)** — broader wallet coverage incl. mobile via QR code
- **Custom SMTP** — Postmark or Resend so invitation emails leave the spam folder
- **Live mainnet** — toggleable mainnet support behind a per-org flag
- **Cross-chain bridge SDK** — replace BRIDGE demo with LayerZero / Wormhole / native bridges
- **USDC / ERC-20 transfers** — token registry + ERC-20 transfer flow
- **Full PQC signing** — replace simulated ML-DSA outputs with real liboqs integration
- **Production HSM** — Fireblocks / Anchorage / BitGo connector option
- **RLS policies** — lock down tables per org_id before non-sandbox use
- **Custodian SAML SSO** — included in Enterprise plan

Sign-off

v3.1 is feature-complete for the institutional sandbox demo. The frontend is production-deployed, the backend is migrated, the blockchain layer is wired to real Sepolia transactions, and the team-onboarding flow works end-to-end. All twenty Vercel review items from the original brief are resolved.

End of report — generated automatically from project changelog.